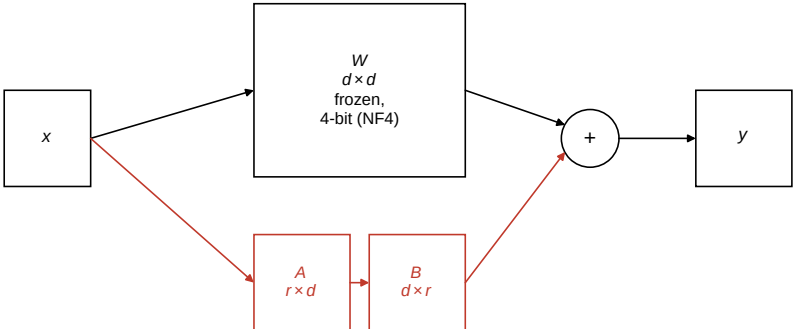
LoRA: freeze the big weight matrix, train a small low-rank update



$$y = Wx + \frac{\alpha}{r} BAx$$

our setting: rank $r = 16$, scale $\alpha = 32$,
adapters on every attention and MLP projection

only A and B are trained;
one adapter is about 42 MB